ZoKrates

A learning group for ZK and SNARK application development

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Respectful collaboration starts here

ALL ARE WELCOME

TODO BIENVENIDOS

みなさん歓迎します。

www.lfdecentralizedtrust.org/code-of-conduct

Agenda

- *Logistics*
- *Zero knowledge proofs - summary*
- *High level flow of zero knowledge programming*
- *ZoKrates language elements*
- *ZoKrates platform*
- *Demo*
- *Challenge*
- *Links, Resources, Literature*
- *Q&A*

Logistics

Every month or two months

Similar structure, similar logistics + ***guest lectures***

Theory:

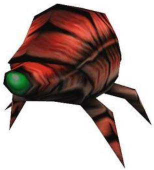
- Look tables,
- Zk variations: STARKs, FRI
- recursive SNARKs

Programming / platforms:

- ZoKrates, Leo
- Zinc / ZKSync

Applications:

- ZK bridges
- ZK Virtual Machines
- ZK Identity / verifiable credential - presentations
- Voting



Discord channel:

https://discord.com/channels/905194001349627914/1329201532628898036

Meetup.com:

https://www.meetup.com/lfdt-hungary/events/305634614/

Repo with all the contents:

https://github.com/LF-Decentralized-Trust-labs/zk-learning-group

Zero knowledge proofs - summary

Computation: arithmetic circuit : *C(x, w) -> F*

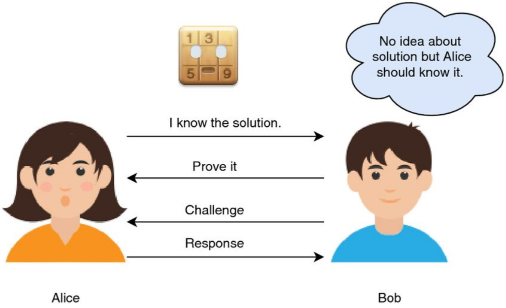
- *x* public input
- *w* private input, witness
- high level computation
- arithmetic circuit
- polynomials

Prover algorithm: *P (pp, x, w) -> proof*

Verifier algorithm: *V (vp, x, proof) -> accept / reject*

Properties:

- *Succinct:*
- *Complete:*
- *Knowledge sound:*
- *Zero knowledge*



High level flow of zero knowledge programming

Computation:

- “High” level programs
- Input-output, program instructions

Arithmetization:

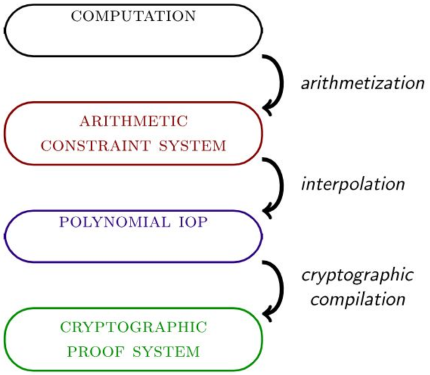
- Arithmetic constraint system
- Equations on finite fields
- Constraint system solution: valid computation

Polynoms

- Polynomial IOP (interactive oracle proof)

Proof system:

- Polynomial commitments



ZoKrates

Toolbox implementing ZkSNARKs

High-level language (DSL), python like language with special elements

ZIR: intermediate representation. Close to R1CS, encapsulating witness generation

Different core algorithms:

- **Groth16**
- **GM17**
- **Marlin**

Tools for trusted setup, generating and verifying proofs

ZoKrates language elements

Compiles to circuits ! *(non-intuitive execution)*

Types:

- Simple types: fields, integers, bool
- Complex types: arrays, tupples, slices, structs

Control structures:

- Conditions, if-then
- Loops (finite, known in compile time)

Building blocks

- Functions
- Generics

ZoKrates platform

Tools for use ZK in real life development scenarios

Compiling, creating proofs, verifying proofs

CLI tools

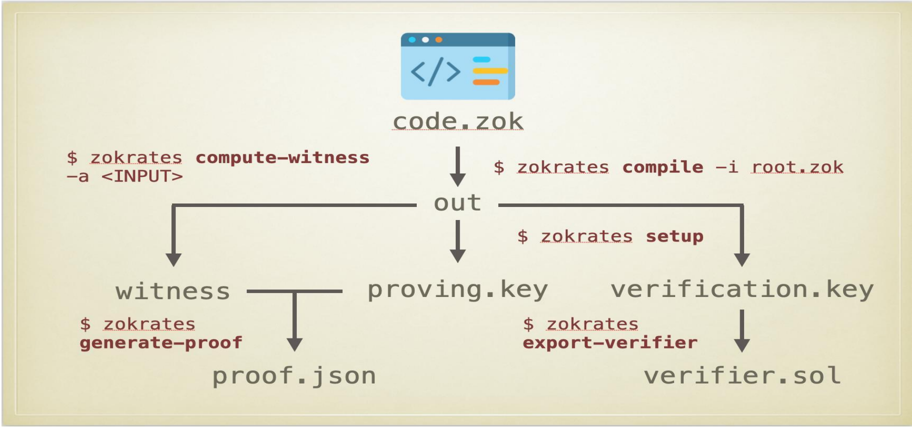
Javascript library: [zokrates.js]

Solidity, Remix integration

Debugging, Uint testing

Standard library: help functions, hash functions, signatures, etc (https://zokrates.github.io/toolbox/stdlib.html)

ZoKrates workflow



Demo

ZK Magic Square

https://zokrates.github.io/examples/magic_square.html

+

Remix Demo

Challenge

Build a simple (9,9) sudoku puzzle with ZoKrates

Links, Resources, Literature

ZoKrates getting started
https://extropy-io.medium.com/zokrates-getting-started-cbcba0c71e4c

ZoKrates — zkSNARKs On Ethereum (made easy)
https://medium.com/coinmonks/zokrates-zksnarks-on-ethereum-made-easy-8022300f8ba6

ZK Study –- ZoKrates
https://hackmd.io/@vivi432/2021-zokrates

ZoKrates documentation
https://zokrates.github.io/

ZKP Zokrates tutorial
https://github.com/robert-zaremba/zkp-zokrates-tutorial

ZoKrates, playground
https://github.com/Zokrates/zokrates-playground

Happy Hunting for the SNARK :)

Q & A

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